Week 6 complete solutions

Problem-sheet solutions

Each solution states the intended method, relevant conditions, derivation, final result, mark allocation and common mistakes. Award method credit after a purely arithmetic slip when the setup and conditions remain valid.

`W06-Q01-PA (M9)`

Method and conditions. For an unconstrained differentiable objective on $\mathbb{R}^2$, an interior local optimum must be stationary. A definite Hessian at a stationary point supplies a strict local sufficient condition.

The column gradient is

$$\nabla f(x, y) = \begin{bmatrix} 2x + y - 6 \\ x + 2y \end{bmatrix}.$$

The stationary equations are

$$2x + y = 6, \qquad x + 2y = 0.$$

They give $y = -2$ and $x = 4$. The Hessian is

$$\mathbf{H}_f = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its leading principal minors are $2 > 0$ and $4 - 1 = 3 > 0$, so it is positive definite.

Final result. The only stationary point is $(4, -2)$. It is a strict local minimum.

Marks (6). Correct column gradient, 1; stationary equations, 1; solution $(4, -2)$, 2; correct Hessian, 1; valid positive-definiteness classification and local conclusion, 1.

Common mistakes. Writing the gradient as a row; solving only one stationary equation; treating the first-order condition as sufficient; misclassifying a positive definite Hessian as a maximum.

`W06-Q01-PB (M9)`

Method and conditions. Separate the necessary first-order condition from the sufficient curvature condition, then check whether the curvature is global on a convex domain.

The equation $\nabla f = \mathbf{0}$ is necessary for an interior differentiable local optimum, but stationary points can be minima, maxima or saddles. It therefore produces candidates rather than a classification.

Here the Hessian is positive definite at every point of the convex domain $\mathbb{R}^2$. Hence f is strictly convex on the whole domain. A stationary point of a differentiable convex function is a global minimiser, and strict convexity makes it unique.

Final result. The point $(4, -2)$ is the unique global minimiser.

Marks (4). Necessary-versus-sufficient distinction, 1; global positive-definiteness or strict-convexity condition, 1; valid global-minimum conclusion, 1; uniqueness, 1.

Common mistakes. Calling every stationary point optimal; using curvature only at the point to assert a global result; omitting the convex domain; reversing convexity and concavity.

`W06-Q02-PA (M9)`

Method and conditions. Solve the interior first-order conditions and use the symmetric Hessian to classify the candidate locally.

$$\nabla g(x,y) = \begin{bmatrix} 8-2x-y \\ 6-x-2y \end{bmatrix}, \qquad \mathbf{H}_g = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}.$$

The stationary equations are

$$2x + y = 8, \qquad x + 2y = 6.$$

Solving gives $x = 10/3$ and $y = 4/3$. The first leading principal minor of $\mathbf{H}_g$ is $-2 < 0$, while its determinant is $4 - 1 = 3 > 0$. Therefore $\mathbf{H}_g$ is negative definite.

Final result. The unique stationary point is $(10/3, 4/3)$, and it is a strict local maximum.

Marks (7). Correct column gradient, 2; stationary equations and solution, 2; correct Hessian, 1; valid negative-definiteness test, 1; local classification, 1.

Common mistakes. Losing a cross-partial term; solving the linear system incorrectly; applying the positive-definite sign pattern; calling the local result global without an additional argument.

`W06-Q02-PB (M9)`

Method and conditions. A negative definite Hessian everywhere on a convex domain makes the objective strictly concave. For a differentiable concave objective, a stationary point is a global maximiser.

The Hessian found in Part A is constant and negative definite throughout $\mathbb{R}^2$, which is convex. Thus g is strictly concave. The stationary point exists and is in the domain, so it is the unique global maximiser.

Its value is

$$\begin{aligned} g\left(\frac{10}{3}, \frac{4}{3}\right) &= \frac{80}{3} + 8 - \frac{100}{9} - \frac{16}{9} - \frac{40}{9} \\ &= \frac{52}{3}. \end{aligned}$$

Final result. The unique global maximiser is $(10/3, 4/3)$, and the global maximum is $52/3$.

Marks (5). Global negative-definiteness condition, 1; convex-domain and strict-concavity statement, 1; valid global and uniqueness conclusion, 1; correct substitution, 1; maximum value, 1.

Common mistakes. Treating local negative definiteness as automatically global; omitting the domain condition; failing to distinguish maximiser from maximum; arithmetic error in the cross term.

`W06-Q03-PA (M9)`

Method and conditions. Find all stationary points, apply the definite or indefinite Hessian tests locally, then study an unbounded path to decide global existence.

$$\nabla h(x,y) = \begin{bmatrix} 3x^2 - 3 \\ -2y \end{bmatrix}, \qquad \mathbf{H}_h(x,y) = \begin{bmatrix} 6x & 0 \\ 0 & -2 \end{bmatrix}.$$

The stationary points are $(-1, 0)$ and $(1, 0)$. At $(-1, 0)$, the Hessian is $\operatorname{diag}(-6, -2)$, which is negative definite, so the point is a strict local maximum. At $(1, 0)$, the Hessian is $\operatorname{diag}(6, -2)$, which is indefinite, so the point is a saddle.

Along the line $y = 0$,

$$h(x, 0) = x^3 - 3x.$$

This tends to $+\infty$ as $x \to +\infty$ and to $-\infty$ as $x \to -\infty$.

Final result. The point $(-1, 0)$ is a strict local maximum and $(1, 0)$ is a saddle. There is no global maximum and no global minimum on $\mathbb{R}^2$.

Marks (6). Correct gradient and both candidates, 2; correct Hessian, 1; two correct local classifications, 2; valid unboundedness argument and both global conclusions, 1.

Common mistakes. Missing one square-root candidate; reversing the Hessian classifications; calling a saddle a minimum; inferring global existence from the finite list of stationary points.

`W06-Q04-PA (M9)`

Method and conditions. Use the first-order condition to locate candidates. When the Hessian test is inconclusive, classify directly from the objective and examine its behaviour on the whole domain.

$$\nabla p(x, y) = \begin{bmatrix} 4x^3 \\ 4y^3 \end{bmatrix}, \qquad \mathbf{H}_p(x, y) = \begin{bmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{bmatrix}.$$

The only stationary point is $(0, 0)$, where $\mathbf{H}_p(0, 0) = \mathbf{0}_{2\times 2}$. The zero matrix is positive semidefinite and negative semidefinite but not definite, so the usual definite-Hessian sufficient tests are inconclusive.

For all (x, y),

$$p(x, y) = x^4 + y^4 \geq 0 = p(0, 0).$$

Equality requires $x = y = 0$. Therefore the origin is the unique global minimiser and hence also a strict local minimiser. As either coordinate tends in magnitude to infinity, $p(x, y) \to \infty$, so p is unbounded above.

Final result. The origin is the unique global minimum, despite the inconclusive Hessian there. No global maximum exists.

Marks (8). Correct gradient and stationary point, 2; correct Hessian and evaluation, 1; explanation of inconclusiveness, 1; direct non-negativity argument, 2; uniqueness and local/global distinction, 1; no-maximum conclusion, 1.

Common mistakes. Calling the zero Hessian positive definite; concluding that an inconclusive test means no optimum; omitting the equality case needed for uniqueness; assuming a stationary point guarantees a global maximum.

`W06-Q05-PA (M9)`

Method and conditions. Classify the constant symmetric Hessian by its eigenvalues, then use the resulting quadratic form to describe global behaviour.

The Hessian is

$$\mathbf{H}_{q_a} = \begin{bmatrix} -2 & -2a \\ -2a & -2 \end{bmatrix}.$$

The directions $[1, 1]^\mathsf{T}$ and $[1, -1]^\mathsf{T}$ have eigenvalues

$$-2(1+a) \quad \text{and} \quad -2(1-a),$$

respectively.

If $|a| < 1$, both eigenvalues are negative. The objective is strictly concave, and $q_a(x, y) < 0$ for every non-zero (x, y), while $q_a(0, 0) = 0$. The origin is the unique global maximiser.

If $a = 1$, then

$$q_1(x, y) = -(x + y)^2.$$

It is concave but not strictly concave, and every point with $y = -x$ is a global maximiser with value zero. If $a = -1$, then

$$q_{-1}(x, y) = -(x - y)^2,$$

and every point with $y = x$ is a global maximiser. In both boundary cases the Hessian is negative semidefinite, and the origin is a non-strict global maximiser.

If $|a| > 1$, the eigenvalues have opposite signs, so the Hessian is indefinite. Along one eigenvector direction the quadratic tends to $+\infty$ after scaling, and along the other it tends to $-\infty$. Thus the origin is a saddle and no global maximum exists.

Final result. Negative definite curvature and a unique global maximum occur for $|a| < 1$; negative semidefinite curvature and a line of global maximisers occur for $|a| = 1$; indefinite curvature, a saddle at the origin and no global maximum occur for $|a| > 1$.

Marks (10). Correct Hessian, 1; correct eigenvalues or equivalent definiteness test, 2; complete $|a| < 1$ classification, 2; both $|a| = 1$ square representations and maximiser sets, 3; complete $|a| > 1$ saddle and unboundedness argument, 2.

Common mistakes. Treating semidefinite as definite; omitting one boundary value of a; reporting only the origin on a ridge of maximisers; claiming an indefinite objective has a global maximum.

`W06-Q06-PA (S6)`

Method and conditions. Apply the course definitions to the realised sample and keep random statistics distinct from their observed values.

$$\bar{x} = \frac{2 + 4 + 4 + 6}{4} = 4.$$

The deviations are $-2, 0, 0, 2$, so

$$s^2 = \frac{(-2)^2 + 0^2 + 0^2 + 2^2}{3} = \frac{8}{3}.$$

Before observation, $\bar{X}$ and S^2 are random variables because they are functions of the random sample. After observing $2, 4, 4, 6$, their realisations are $\bar{x} = 4$ and $s^2 = 8/3$.

Final result. $\bar{x} = 4$ and $s^2 = 8/3$.

Marks (5). Correct mean, 1; deviations and squared-deviation sum, 1; correct $n - 1$ denominator, 1; correct variance, 1; random-versus-observed distinction, 1.

Common mistakes. Dividing the variance sum by n; using deviations from zero; failing to square negative deviations; calling observed values random variables.

`W06-Q06-PB (S6)`

Method and conditions. For an i.i.d. normal sample with population variance σ^2, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ exactly. Here $n = 10$ and $\sigma^2 = 4$, so

$$\frac{9S^2}{4} \sim \chi^2_9.$$

Moreover,

$$\mathbb{P}(S^2 \le c) = \mathbb{P}\left(\frac{9S^2}{4} \le \frac{9c}{4}\right).$$

To make this probability 0.95, set $9c/4 = 16.919$. Hence

$$c = \frac{4(16.919)}{9} = \frac{67.676}{9} \approx 7.52.$$

The exact chi-squared sampling result requires that the common population distribution be normal, in addition to the stated i.i.d. sampling.

Final result. $9S^2/4 \sim \chi^2_9$, and $c = 4(16.919)/9 \approx 7.52$.

Marks (8). Correct chi-squared statistic, 2; correct degrees of freedom, 1; probability transformation, 2; equation for and solution of c, 2; normality condition, 1.

Common mistakes. Using n rather than $n-1$; treating 4 as a standard deviation; inverting the chi-squared scaling; omitting the population normality condition.

`W06-Q07-PA (S6)`

Method and conditions. Independence and normality make both linear combinations exactly normal. Use the variance, rather than the standard deviation, as the second normal parameter.

$$\bar{X} \sim \mathcal{N}\left(20, \frac{16}{16}\right) = \mathcal{N}(20, 1),$$

and

$$T \sim \mathcal{N}(16 \cdot 20, 16 \cdot 16) = \mathcal{N}(320, 256).$$

Thus

$$\begin{aligned}\mathbb{P}(18 < \bar{X} < 22) &= \mathbb{P}(-2 < Z < 2) \\ &= 2\Phi(2) - 1 \\ &= 2(0.9772) - 1 = 0.9544.\end{aligned}$$

The standard deviation of T is 16, so

$$\mathbb{P}(T > 336) = \mathbb{P}(Z > 1) = 1 - \Phi(1) = 0.1587.$$

Final result. The exact distributions are $\bar{X} \sim \mathcal{N}(20, 1)$ and $T \sim \mathcal{N}(320, 256)$. The probabilities are 0.9544 and 0.1587, respectively.

Marks (7). Correct distribution of $\bar{X}$, 2; correct distribution of T, 2; first standardisation and probability, 2; second standardisation and probability, 1.

Common mistakes. Using 16 as the population standard deviation; dividing the variance of the sum by n; standardising T with the standard error of $\bar{X}$; treating the second normal parameter as a standard deviation.

`W06-Q07-PB (S6)`

Method and conditions. The observations are i.i.d. with finite positive variance, so the central limit theorem supports a large-sample normal approximation for the standardised sample mean.

For a Bernoulli(0.5) variable,

$$\mu = 0.5, \qquad \sigma^2 = 0.5(1 - 0.5) = 0.25.$$

Therefore

$$\mathrm{Var}(\bar{Y}) = \frac{0.25}{100} = 0.0025, \qquad \mathrm{SD}(\bar{Y}) = 0.05.$$

The standardised endpoints are

$$\frac{0.4 - 0.5}{0.05} = -2, \qquad \frac{0.6 - 0.5}{0.05} = 2.$$

Hence, without a continuity correction,

$$\mathbb{P}(0.4 \leq \bar{Y} \leq 0.6) \approx \Phi(2) - \Phi(-2) = 2\Phi(2) - 1 = 0.9544.$$

This is approximate because a Bernoulli sample mean has a discrete distribution and is not exactly normal. The CLT supplies the normal approximation at large n.

Final result. The requested CLT approximation is 0.9544.

Marks (7). Correct Bernoulli mean and variance, 1; correct variance or standard deviation of $\bar{Y}$, 2; two correct standardisations, 2; probability calculation, 1; exact-versus-approximate explanation, 1.

Common mistakes. Using 0.25 as the standard deviation; omitting the $\sqrt{n}$ scaling; claiming exact normality; standardising the observations instead of the sample mean.

`W06-Q08-PA (S6)`

Method and conditions. Substitute the common finite variance and the specified tolerance into the supplied concentration bound, then enforce the integer sample-size condition.

$$\mathbb{P}(|\bar{X} - 50| \geq 1) \leq \frac{25}{n}.$$

We require

$$\frac{25}{n} \leq 0.04 = \frac{1}{25}.$$

Since $n > 0$, this is equivalent to $n \geq 625$. Thus the smallest integer is 625.

As $n \to \infty$, the bound $25/n$ tends to zero. This supports $\bar{X} \xrightarrow{p} 50$, the law-of-large-numbers conclusion. The bound concerns concentration and does not specify a normal sampling shape.

Final result. The smallest sample size certified by the bound is $n = 625$.

Marks (6). Correct substitution into the bound, 1; correct inequality, 1; correct algebra, 1; integer boundary, 1; convergence-in-probability interpretation, 1; no-normality distinction, 1.

Common mistakes. Using the standard deviation in place of the variance; reversing the sample-size inequality; rounding below the boundary; claiming the bound proves approximate normality.

`W06-Q08-PB (S6)`

Method and conditions. State the limiting object and convergence mode for each result under the stated i.i.d. and finite-variance framework.

The law of large numbers says

$$\bar{X} \xrightarrow{p} \mu.$$

Equivalently, for every $\varepsilon > 0$, $\mathbb{P}(|\bar{X} - \mu| > \varepsilon) \to 0$. It does not say that $\bar{X}$ equals μ at a finite sample size or that it becomes normal.

For $0 < \sigma^2 < \infty$, the central limit theorem says

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

It supplies an approximate sampling shape for the mean at large n, with scale $\sigma/\sqrt{n}$. Neither theorem says that the individual observations become normal as the sample size grows.

Final result. The law of large numbers concerns concentration of $\bar{X}$ at μ; the central limit theorem concerns the limiting distribution of the correctly standardised sampling error.

Marks (5). Correct LLN convergence and object, 1; finite-sample qualification, 1; correct CLT standardisation, 1; distributional conclusion, 1; correct statement about individual observations, 1.

Common mistakes. Describing the LLN as a normal approximation; omitting the $\sqrt{n}$ scaling; asserting finite-sample equality; claiming the observations themselves become normal.

`W06-Q09-PA (M6; M7 retrieval)`

Method and conditions. Apply the course convention that gradients are column vectors. Since F is a polynomial, it is continuously differentiable. To define $x = x(\theta)$ locally, the relevant implicit-function condition is $F_x \neq 0$.

At the stated point,

$$F(1, 2) = 1 + 2 - 3 = 0.$$

The partial derivatives and column gradient are

$$F_x(x, \theta) = 2x + \theta, \qquad F_\theta(x, \theta) = x, \qquad \nabla F(x, \theta) = \begin{bmatrix} 2x + \theta \\ x \end{bmatrix}.$$

Thus

$$\nabla F(1,2) = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \qquad \dim(\nabla F) = 2 \times 1.$$

Moreover, $F_x(1,2) = 4 \neq 0$, so the equation defines a local branch $x = x(\theta)$ through the point. Along that branch,

$$0 = \mathrm{d}F = F_x \,\mathrm{d}x + F_\theta \,\mathrm{d}\theta,$$

and hence

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\frac{F_\theta}{F_x} = -\frac{x}{2x+\theta}.$$

Final result. At $(x,\theta) = (1,2)$,

$$\left.\frac{\mathrm{d}x}{\mathrm{d}\theta}\right|_{(1,2)} = -\frac{1}{4}.$$

Marks (5). Verification that the point satisfies the equation, 1; correct column gradient and dimension, 2; valid local implicit-function condition, 1; correct total-differential derivative and evaluation, 1.

Common mistakes. Writing the gradient as a row; checking $F_\theta \neq 0$ instead of $F_x \neq 0$; omitting the minus sign; reporting the general derivative without evaluating it at the stated point.

`W06-Q10-PA` **(`M8` retrieval)**

Method and conditions. Represent the quadratic form by a symmetric matrix and apply the leading-principal-minor test.

$$k(u,v) = \begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}.$$

The leading principal minors are

$$2 > 0, \qquad \det \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = 6 - 1 = 5 > 0.$$

Thus the symmetric matrix is positive definite. The Hessian of k is twice this matrix and is also positive definite everywhere.

Final result. The quadratic form is positive definite, and k is strictly convex on $\mathbb{R}^2$.

Marks (5). Correct symmetric matrix, 2; first principal minor, 1; determinant, 1; definiteness and curvature conclusions, 1.

Common mistakes. Using 2 rather than 1 in each off-diagonal entry; checking only the determinant; confusing positive definite with positive semidefinite; inferring concavity instead of convexity.

`W06-Q11-PA` **(`S3`; `S4`; `S5` retrieval)**

Method and conditions. Use the stated exponential rate parameterisation, then use independence to obtain conditional moments. Apply both iterated expectation and the law of total variance.

$$\mathbb{E}[X] = \frac{1}{2}, \qquad \mathrm{Var}(X) = \frac{1}{4}.$$

Since X and B are independent,

$$\mathbb{E}[Y \mid B] = 3\mathbb{E}[X \mid B] + B = \frac{3}{2} + B.$$

This random variable takes the value $3/2$ when $B = 0$ and $5/2$ when $B = 1$. By iterated expectation,

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid B]] = \frac{3}{2} + \mathbb{E}[B] = 2.$$

Conditional on B, the variable B is constant. Therefore

$$\operatorname{Var}(Y \mid B) = 9 \operatorname{Var}(X \mid B) = \frac{9}{4}.$$

Also,

$$\operatorname{Var}(\mathbb{E}[Y \mid B]) = \operatorname{Var}\left(\frac{3}{2} + B\right) = \operatorname{Var}(B) = \frac{1}{4}.$$

The law of total variance gives

$$\operatorname{Var}(Y) = \mathbb{E}[\operatorname{Var}(Y \mid B)] + \operatorname{Var}(\mathbb{E}[Y \mid B]) = \frac{9}{4} + \frac{1}{4} = \frac{5}{2}.$$

Final result. $\mathbb{E}[Y \mid B] = 3/2 + B$, $\mathbb{E}[Y] = 2$, and $\operatorname{Var}(Y) = 5/2$.

Marks (6). Correct exponential moments, 1; correct conditional-expectation random variable, 1; valid iterated-expectation setup and mean, 2; both total-variance components and final variance, 2.

Common mistakes. Treating the exponential rate as a mean; omitting conditioning on B; averaging conditional means without the Bernoulli probabilities; omitting one component of total variance.

Self-test solutions and error check

Solutions and error check

Solution to `W06-ST01-PA (M9)` Method. Solve the first-order conditions, classify with the Hessian, and use global strict concavity on the convex domain.

$$\nabla f(x, y) = \begin{bmatrix} 4 - 2x \\ 2 - 4y \end{bmatrix}, \qquad \mathbf{H}_f = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}.$$

The stationary point is $(2, 1/2)$. The Hessian is negative definite everywhere, so f is strictly concave on $\mathbb{R}^2$. Hence this point is both a strict local maximum and the unique global maximum. Its value is

$$f\left(2, \frac{1}{2}\right) = 8 + 1 - 4 - \frac{1}{2} = \frac{9}{2}.$$

Marks (5). Correct gradient and stationary point, 2; Hessian and local classification, 1; strict-concavity global justification, 1; maximum value, 1.

Common mistakes. Solving only one first-order condition; reversing the negative-definite classification; asserting a global result without whole-domain curvature; omitting the maximum value.

Solution to `W06-ST02-PA (M9)` Method. Recognise that the Hessian test is inconclusive and inspect directions through the origin.

$$\nabla g(x,y) = \begin{bmatrix} 4x^3 \\ -4y^3 \end{bmatrix}, \qquad \mathbf{H}_g(x,y) = \begin{bmatrix} 12x^2 & 0 \\ 0 & -12y^2 \end{bmatrix}.$$

Both the gradient and Hessian are zero at the origin, so the definite-Hessian test is inconclusive. Along $y = 0$, $g(x,0) = x^4 > 0$ for $x \neq 0$, whereas along $x = 0$, $g(0,y) = -y^4 < 0$ for $y \neq 0$. Every neighbourhood of the origin contains values above and below zero, so the origin is a saddle. The same two paths show that g is unbounded above and below.

Marks (5). Correct gradient and stationary point, 1; correct Hessian and inconclusive result, 1; positive direction, 1; negative direction and saddle classification, 1; no global maximum or minimum, 1.

Common mistakes. Calling the zero Hessian definite; treating an inconclusive test as a minimum; checking only one path; claiming a global optimum exists.

Solution to `W06-ST03-PA (S6)` Method. Use the course definitions with denominator $n-1$ for the sample variance.

$$\bar{x} = \frac{1+3+5+7}{4} = 4.$$

The deviations are $-3, -1, 1, 3$, so

$$s^2 = \frac{9+1+1+9}{3} = \frac{20}{3}.$$

The quantities $\bar{X}$ and S^2 are random statistics before the sample is observed; $\bar{x} = 4$ and $s^2 = 20/3$ are their observed values.

Marks (5). Correct mean, 1; deviations and squared sum, 1; denominator, 1; variance, 1; random-versus-observed distinction, 1.

Common mistakes. Dividing by 4; taking deviations from zero; failing to square negative deviations; using upper-case notation for the observed values.

Solution to `W06-ST04-PA (S6)` Method. Use the exact normal sampling distribution and standardise with the standard deviation of the sample mean.

$$\bar{X} \sim \mathcal{N}\left(5, \frac{9}{9}\right) = \mathcal{N}(5,1).$$

Therefore

$$\begin{aligned} \mathbb{P}(4 < \bar{X} < 6) &= \mathbb{P}(-1 < Z < 1) \\ &= 2\Phi(1) - 1 \\ &= 2(0.8413) - 1 \\ &= 0.6826. \end{aligned}$$

Marks (5). Correct mean of $\bar{X}$, 1; correct variance, 1; exact normal distribution, 1; standardisation, 1; probability, 1.

Common mistakes. Using variance 9 for $\bar{X}$; treating 9 as a standard deviation; invoking the CLT instead of exact normality; omitting symmetry in the two-sided probability.

Record the first cause of each error under `M9` or `S6`, then reattempt that part from a blank page in two or three days.